ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. 594
2. Because different categories have different characteristics, applying an algorithm to mixed data would fail to capture each category's specific characteristics.
   Thus, it is necessary to isolate individual categories first before applying algorithms to each of them.

AI used for this answer: Yes

If yes, how: I asked AI what frequent pattern mining is, and asked it to translate my answer.

External references (links):

https://www.geeksforgeeks.org/dsa/frequent-pattern-mining-in-data-mining/

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer:

Discarding 40% of the data would make it harder to detect meaningful characteristics.

Instead of dropping these rows, the missing category_name can be recovered by mapping it back from its corresponding category numeric value. If the category value itself is missing, it may be necessary to train a classifier to infer the category from the text content.

AI used for this answer: Yes

If yes, how: I asked AI to correct the grammar of my answer.

External references (links):

https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.dropna.html

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Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. No, the proportions aren't the same.
2. When sampling from a highly imbalanced dataset, categories with a low proportion of the dataset may not be drawn into the sampled subset.

AI used for this answer: Yes

If yes, how: I asked AI to check the grammar.

External references (links):

https://developers.google.com/machine-learning/crash-course/overfitting/imbalanced-datasets?hl=zh-tw

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Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?

2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. The total number of features (columns) increase.
   stop_words='english'：(2257, 35482)
   stop_words=None：(2257, 35788)

2. The overall sparsity of the matrix would decrease, because stopwords appear in nearly every text, compared to other words that may only appear in some documents. Adding back these words lowers the proportion of zeros in the matrix, thus decreasing its sparsity.

AI used for this answer: Yes

If yes, how: I asked AI to check the grammar of the answer.

External references (links):

https://en.wikipedia.org/wiki/Sparse_matrix

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

1.

threshold	kept	removed
0.0	35482	0
0.01	5089	30393
0.05	1491	33991

As the threshold increases from 0.0 to 0.05, the total size of kept vocabulary decreases. When we set the threshold from 0.0 to 0.01, we removed 30393 words, but when we set threshold from 0.01 to 0.05, it removed 3598 more words.

2.

The words that have low document frequency have variance close to zero and would be eliminated first.

3.

Words with near-zero variance are typically typos or rarely appear across the documents, giving them no discriminative power. Thus, they can be eliminated without raising any accuracy concerns.

Threshold	Number of terms kept (keep == True)
0.0	35482
0.01	5089
0.05	1491

AI used for this answer: Yes

If yes, how: I asked it what document frequency is and let it polish the grammar of my answer.

External references (links):

https://medium.com/willhanchen/%E8%87%AA%E7%84%B6%E8%AA%9E%E8%A8%80%E8%99%95%E7%90%86-%E6%A6%82%E5%BF%B5%E7%AF%87-%E6%8E%A2%E7%B4%A2tf-idf-%E9%97%9C%E6%96%BC%E8%A9%9E%E7%9A%84%E7%B5%B1%E8%A8%88%E8%88%87%E7%B4%A2%E5%BC%95%E9%9A%B1%E5%90%AB%E8%91%97%E4%BB%80%E9%BA%BC%E5%A5%A7%E7%A7%98%E5%91%A2-6e0079151d0a

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Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

	feature	pearson_r	pearson_p
0	thanks	0.277827	2.782173e-41
1	writes	-0.233682	2.270527e-29
2	keywords	0.232965	3.395522e-29
3	looking	0.213468	1.133072e-24
4	vga	0.211099	3.755895e-24
5	video	0.209458	8.545074e-24
6	people	-0.208871	1.144445e-23
7	card	0.205532	5.934532e-23
8	hi	0.202522	2.553534e-22
9	host	0.199737	9.655856e-22
10	algorithm	0.195873	5.916113e-21
11	nntp	0.191793	3.852599e-20
12	graphics	0.177855	1.698002e-17
13	rutgers	-0.176603	2.865961e-17
14	algorithms	0.176227	3.350960e-17

	feature	spearman_r	spearman_p
0	graphics	0.521069	2.446204e-157
1	people	-0.319349	1.115413e-54
2	files	0.316719	9.169093e-54
3	image	0.308622	5.266545e-51
4	god	-0.307345	1.408580e-50
5	3d	0.305991	3.975746e-50
6	program	0.291122	2.477587e-45
7	thanks	0.288667	1.440824e-44
8	file	0.283677	4.890639e-43
9	windows	0.276869	5.346111e-41
10	ftp	0.274446	2.750797e-40
11	format	0.266011	7.263356e-38
12	gif	0.258214	1.054474e-35
13	package	0.256314	3.458413e-35
14	writes	-0.256210	3.689453e-35

1.

Both generic stop words and domain-specific graphics terminology appear in the top 15 for both methods. However, Pearson's top ranks are dominated more by generic words (e.g., "thanks", "writes"), while Spearman's top ranks are dominated more by domain-specific terms (e.g., "graphics", "image").

2.

Pearson correlation measures the strength of a linear relationship between two variables, while Spearman correlation measures the strength of a monotonic relationship by assessing the consistency of their rank ordering.

3.

The word "graphics" ranks 1 in Spearman but only rank 13 in Pearson. This is likely because the word doesn't appear evenly. Most comp.graphics articles may mention it once or twice, but some may mention it many times. Spearman only cares about the order, not the exact numbers. The more it's mentioned in comp.graphics articles, the higher it ranks in Spearman. But Pearson is sensitive to outliers (articles that mention it many times), so the uneven counts weaken its linear correlation.

AI used for this answer: Yes

If yes, how: I asked it about the principles of Spearman and Pearson, then asked it to polish the grammar of my answer.

External references (links):

https://flag-editors.medium.com/%E6%A9%9F%E5%99%A8%E5%AD%B8%E7%BF%92%E5%8B%95%E6%89%8B%E5%81%9Alesson-8-%E8%88%87%E8%81%B7%E5%A0%B4%E6%81%AF%E6%81%AF%E7%9B%B8%E9%97%9C%E7%9A%84pearson-spearman-kendall%E7%9B%B8%E9%97%9C%E4%BF%82%E6%95%B8-%E4%B8%8A%E7%AF%87-87c93d38f27e

https://en.wikipedia.org/wiki/Spearman%27s_rank_correlation_coefficient

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Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Top-K FAE

k	runtime	the last one
100	0.11851143836975098	99 sea bronx 21
500	0.322674036026001	499 sank queens manhattan blew beaverton 16

MaxFPGrowth

minSup	runtime	total number of discovered patterns
9	0.1752614974975586	710
3	3.784578323364258	9547

Answer (1) — Runtime and pattern count when lowering minSup:

When we decrease minSup from 9 to 3, the execution runtime increases from 0.175 to 3.785, and the total number of discovered patterns increases from 710 to 9547.

So the runtime and total number of patterns increase when the minSup decreases.

Answer (2) — Top-K k vs MaxFPGrowth minSup:

The parameter k determines how many patterns to extract, regardless of their frequency. minSup, on the other hand, sets a frequency threshold, regardless of how many patterns end up passing it.

Answer (3) — Maximal vs standard frequent patterns:

Because maximal frequent patterns only keep the largest itemsets and discard all their subsets, the total count is much smaller than standard frequent patterns, which keep every subset as well.

AI used for this answer: Yes

If yes, how: Discussed the content of Section 5.4, summarized key points, had it explain maximal vs. standard frequent patterns, and corrected the grammar of my answers.

External references (links):

https://www.geeksforgeeks.org/data-science/frequent-item-set-in-data-set-association-rule-mining/

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1.

Mathematically, in high-dimensional space, data becomes very sparse, causing the distances between data points to become nearly identical, which makes it impossible to meaningfully group them by distance.

Visually, humans can only perceive two or three dimensions.

2.

Dimensionality reduction algorithms like PCA, t-SNE or UMAP should be applied first.

AI used for this answer: Yes

If yes, how: Had it corrected the grammar of my answers.

External references (links):

https://biostatsquid.com/pca-umap-tsne-comparison/

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1.

UMAP shows distinct, separated cluster islands. t-SNE has some points overlapping around the center. PCA shows the heaviest overlap, with nearly all points clustered so tightly that they look like a single point.

2.

Because high-dimensional data often doesn't have a linear structure, and PCA is a linear dimensionality reduction technique, it is less effective at capturing non-linear relationships and easily influenced by outliers.

3.

t-SNE distorts distances in order to preserve the local structure between data points, so the distance between clusters in a t-SNE plot cannot be used to judge how different those clusters actually are.

AI used for this answer: Yes

If yes, how: Explained the principles of PCA, t-SNE, and UMAP, and corrected grammar.

External references (links):

https://www.kaggle.com/code/samuelcortinhas/intro-to-pca-t-sne-umap

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Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1.

4 columns are created to represent the 4 categories.

2.

The algorithm might mistakenly assume that categories with closer numbers are more similar, while categories with larger numerical differences are less similar.

For example, it might treat category 0 and 1 as more similar than category 0 and 3.

3.

Through one-hot encoding, every category vector has the same length, and the distance between any two categories is equal, which eliminates the false assumption.

AI used for this answer: Yes

If yes, how: Corrected the grammar of my answer.

External references (links):

https://www.geeksforgeeks.org/machine-learning/ml-one-hot-encoding/
